Research memo: the arithmetic-geometric direction

PRIMES-Conjectures project — AG track, branch `arith-geom-interpretation` — October 21–22, 2026
Full development: `doc/arith-geom.tex`; standalone Popovych note: `doc/popovych-note.tex`;
rebuilt audit PDFs: `doc/audit-pdfs/`; results log: `doc/arith-geom-findings.md`

Audit revision (22 October 2026). The algebraic-group viewpoint gives useful exact identities and containing support bounds, but it does not turn N1/A10 into impossibility theorems. Necessary smoothness requirements use section-specific support hypotheses. The full development and Popovych note have been corrected accordingly; the same-name PDFs beside those sources are historical. Rebuilt versions are in `doc/audit-pdfs/`; current status is in `doc/integration-review-status.md`.

1. The dictionary

The AKS congruence $(X-1)^n \equiv X^n - 1 \pmod{n, X^r-1}$ is a *Frobenius-lift test*: $\mathbb{Z}[X]/(X^r-1)$ is the Λ-ring of μ_r with Adams operations $\psi^k(X) = X^k$. Prime n implies $\psi^n(f) \equiv f^n \pmod n$, but the converse fails in a fixed truncated ring: $399001 = 31 \cdot 61 \cdot 211$, $r = 5$, passes the coincidence on every section, inside the exception clause. Wilkerson's criterion does not assert that converse. AKS checks one section. The exception clause $n^2 \equiv 1 \pmod r$ says exactly that the fake Frobenius σ_n lies in $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+) = \{1, \mathrm{conj}\}$ — the subgroup $\{1, \mathrm{conj}\}$. Choosing a comparable exception for an elliptic test is a modeling decision, not forced by this dictionary.

On the counterexample side, the Lenstra–Pomerance conditions $p-1 \mid n-1$, $p+1 \mid n+1$ say that *n impersonates the Frobenius of p on both one-dimensional algebraic tori over $\mathbb{F}_p$* (split: order $p-1$; nonsplit norm-one: order $p+1$). Twin-smoothness of $(\frac{p-1}{2}, \frac{p+1}{2})$ is torus-order smoothness — the same condition behind the B-SIDH/SQIsign parameter hunts, which is why the project's reliance on the Costello–Naehrig corpora was never a coincidence.

2. The five results

(AG2) Exact support bound; conditional smoothness model. For $d = \mathrm{ord}_r(p)$, the unique involution of $(\mathbb{Z}/r)^\times$ being -1 forces the exact *inversion collapse* $(X-1)^{p^{d/2}-1} = -X^{-1}$ (even d), so the control order $\rho_r(p)$ lives on cyclotomic levels $e \mid d'$, $d' = d/2$ or d, and the containing torus has dimension $\sum_{e \mid d'} \varphi(e) = d'$. Validity ($\mathrm{ord}_r(n) > 2$) forces $d \geq 3$, so **the containing model has dimension ≥ 2, with the minimum 2 attained exactly at $d = 4$, levels $\{1, 2\}$: the double (twin-smooth) condition** — realizable iff $4 \mid r-1$, and Váňa's $\rho_5(p) \mid 10(p^2-1)$ is the special case. The burden formula reproduces the A11 empirical table at all 11 moduli (value and minimizing d) in the historical table. Converting the bound into a necessary B-smoothness condition requires every prime above B in the containing product to occur in the actual order. Neither the density of failures nor the cost of targeting them is established. No modulus direction is closed.

(AG5) Conditional comparison for Popovych's pair. The first section has a torsion cofactor under inversion. The second Popovych section $X+2$ has trivial global torsion-ratio stabilizer: $(\zeta^\gamma+2)/(\zeta+2)$ is not even a unit of $\mathbb{Z}[\zeta_r]$ for $\gamma \neq 1$ (disjoint ideal supports: norms r vs $(2^r+1)/3$). This rules out a global torsion-cofactor identity, not special reductions: at $(r, p) = (13, 5)$, $(X+2)^{24} \in \mu_{26}$. With a separate support-sharpness hypothesis the pair has model dimension burden $\min\{d : d \mid r-1, d \geq 3\} \geq 3$, equal to 4 at Popovych's own $r = 5$: pool primes would need $p-1$, $p+1$ *and* p^2+1 simultaneously smooth. At the frozen harvest spec (pool primes $\approx$51 bits) that extra $\approx$102-bit level is a Dickman tax $\rho(102/\log_2 B)$: 3.9×10^{-11} over the solver base $B = 1259$, still 1.8×10^{-7} over the generous $B' = 12590$ in the polynomial-smoothness heuristic, not a universal harvest bound. Bonus structure: the second congruence's $X{=}1$ component is exactly the base-3 Fermat test $3^n \equiv 3$, and mod 3 the two sections coincide.

(AG5b, Oct 22 addendum) Does the "2" matter? No — but "not 1" is subtly incomplete. For any integer $c \notin \{0, \pm 1\}$ the global argument applies to $X + c$ with the stated exclusions; a necessary smoothness jump additionally needs that section's support hypothesis, with c entering through the norm $\Phi_r(-c) = (c^r+1)/(c+1)$ (finitely many excluded p, with $p \mid c(c+1)$), the excluded modulus $r \mid c+1$, and the level-one component — the base-$(c+1)$ Fermat test. What must be avoided is exactly degeneracy of the section's zero $-c$: *it must be a point of $\mathbb{G}_m$ avoiding the rational torsion* $\{\pm 1\}$, equivalently $c \notin \{0, \pm 1\}$. Here $c = 0$ is excluded trivially, not subtly: X is a monomial, $(X)^n \equiv X^n$ is an identity in every ring (the zero sits at the origin, off the torus, and the section is a unit of finite order r). Among nonzero c: $c = -1$ is the first section itself, and the innocent-looking $c = +1$

fails *outright*. Its zero -1 is inversion-fixed, so the collapse survives ($\mathrm{ord}(X{+}1) \mid r(p^{d/2}{-}1)$ at even d, measured 22/22); worse, $\zeta{+}1 = (\zeta^2{-}1)/(\zeta{-}1)$ with both numerator and denominator values of $X-1$ at primitive roots, whose orders divide $\rho_r(p)$ — so the mechanism congruences plus the Carmichael condition imply $T(+1, n, r)$ *automatically* (proved: the $X+1$ proposition in the note and full document; historical checks: 48/48 synthetic mechanism instances pass $T(+1)$, 0/48 pass $T(+2)$). The hypothetical "$\{X{-}1, X{+}1\}$ conjecture" — Popovych's statement with 2 replaced by 1 — is therefore *mechanism-equivalent* to Agrawal's: formally stronger-looking, but every counterexample the known engine could ever build defeats both simultaneously. In stabilizer terms $U(X \pm 1) = \{\pm 1\}$ while $U(X + c) = \{1\}$ otherwise: $c = 2$ is simply the smallest positive nondegenerate constant in this comparison ($c = -2$ works equally, with Mersenne norm $2^r - 1$).

(AG3) The impersonation hierarchy — our targets have a name, and the wall has a shape. Define cells by (level set, Frobenius powers); at level 2 the power's parity is a sign. Then: Carmichael = $(\{1\}, 0)$, infinite (AGP); Lucas–Carmichael = $(\{2\}, 1)$, infinite (Wright 2018); the L–P targets = $(\{1, 2\}, (0, 1))$ are the *Williams numbers* (Williams 1977, Echi) — open even for nonemptiness, none below 2^{64} (McIntosh–Mitra 2015). Yet Howe's *rigid order-2 Carmichael number* 443,372,888,629,441 = 17·31·41·43·89·97·167·331 occupies the *same level set* with all-even parities and exists at 4.4×10^{14} (re-verified). **Sign splitting is a useful distinction, not a proved wall**: rigid cells are one-sided AGP problems (single target 1); the Williams parities $(0, 1)$ force targets $(+1, -1)$ that conflict on shared prime powers, forcing the coprime side-split studied by N1/A10. Those finite experiments do not prove unconstructibility. The original full-bound $r = 5$ design requires an odd matching exponent; sharp datum: Howe's $n \equiv 1 \pmod 5$ does not even pass $T(-1, n, 5)$ — rigidity misses the μ_{2r} torsion the mod-80 conditions exist to fix.

(AG4) A stabilizer-index upper bound, not elliptic closure. The chosen commuting endomorphism-lift formulation restricts to CM curves; it does not exclude all other elliptic tests. Gurney's thesis treats CM moduli and curves of Shimura type with the relevant base hypotheses. On an ordinary split eigenline Frobenius acts on an r-eigenline by a scalar λ, curve automorphisms give exact collapses (e.g. $x([i]S) = -x(S)$ on $y^2 = x^3 - x$), and everything unifies:

$$\mathrm{ord}(f(T)) \mid L(p^{k'} - 1), \qquad k' = [\langle\lambda\rangle : \langle\lambda\rangle \cap U],$$

L a bound on the cofactor orders, and U the cofactor subgroup ($\{\pm 1\}$ for $X{-}1 \to$ AG2; $\{1\}$ for $X{+}2 \to$ AG5; μ_4/μ_6 for CM x-coordinates). The tantalizing index-1 cells ($\lambda \equiv \pm i$) lie in the chosen exception subgroup. Index-2 cells have a twin-smooth *containing* support; necessity would require the extra saturation hypothesis. The historical computations on $y^2 = x^3 - x$ reported model floors 3 ($r = 13$) and 2 ($r = 17$), matching the index formula; elliptic $\geq$ torus model dimension at all 11 listed moduli. These do not prove support saturation, the necessity of the chosen exception, or transfer of a solver obstruction. The elliptic direction remains open.

3. Epistemic status

Exact: the torus inversion identity, its order upper bound, the characteristic-zero non-unit argument, the stated $X + 1$ implication, and the cofactor-index upper bound under its explicit assumptions.

Conditional: necessary smoothness requirements, assuming prime support at the chosen threshold for each section. A cyclic-group calculation uses $\ell^{v_\ell(N)}$-th powers, not just ℓ-th powers; algebraic dependencies prevent assuming independent uniform reductions without proof.

Unestablished: a density theorem for exceptional orders, production-scale harvest/solver costs, universal elliptic closure, and the proposed function-field sharpness and smooth-order counting theorems. Weil's bounds for specified extensions do not establish those counting statements without extensions, asymptotic regimes and uniform error estimates. Historical committed JSON files are finite checks, not proofs of these claims.

4. Where this leaves the program

- **Historical negative-result paper**: remains on the separate `negative-result-writeup` branch. It is not merged or revised by this audit. Its theorem and infeasibility claims require independent correction.
- **Standalone Popovych note**: the current source distinguishes global identities, finite-field exceptions, conditional support and heuristic cost. It is still a working note, not certified ready for publication.

- **Constructive work**: validate the real pool's units, prime-power moduli, target reachability, parity and component distribution before scaling. A8/A10 forced-zero parameters are not A12 literal identity probabilities; the quoted $\sim$0.004 is not a verified full H7 component-density measure. No single surviving solver paradigm has been proved to be the only option.
- **Open problems surfaced**: unconditional sharpness (equidistribution); existence of Williams numbers at all (independent interest; the sign-split lens suggests why the AGP machine hasn't reached them); whether any side-separating polynomial family reaches an admissible local cell (old Q6.2, still open).